Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/20/2006
PI Name:	Lathan, Corinna E. Ph.D.		
Project Title:	Mental Representation of Spatial Cues (3D-Space)		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20910	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	ILSRA 2001
Start Date:	05/02/2005	**End Date:**	09/30/2010
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	Unassigned Co-Investigator to ESA managed flight study that is in flight development phase.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clement, Gilles (CNRS (Centre national de la recherche scientifique))		
Grant/Contract No.:	NNJ05HD92A		
Performance Goal No.:			
Performance Goal Text:			
	The purpose of this study is to investigate the effects of acute and long-term effects of exposure to microgravity on the mental representation of spatial cues by astronauts during and after space flight. Previous space studies carried out by our group have shown that cognitive functions such as mental rotation, symmetry detection, and the internal representation of the vertical axis of objects were found to be impaired during and after space flight. We propose to repeat the preliminary observations on a larger number of human subjects during short-and-long-term space missions, by using simple experiments requiring a laptop computer and a digitizing table. The analysis of the spatial and dynamic components of writing and drawing tests of 2-D and 3-D objects performed throughout a space mission should allow to confirm that the internal spatial representation of the vertical dimension, which is determined on Earth by the gravitational reference system, is altered during space flight. If crew time and equipment availability permit, additional psychophysics measurements will be made using a portable virtual reality system displaying 3-D objects with one		

Task Description:	dimension altered. Tests of the perception of absolute distances will also be performed using the same equipment. The results of this study could have important consequences for human performance during space flight. For example, if an astronaut imagines a space station shorter than it is in reality, the expected duration of a trip within this station will be shorter than what it actually takes, thus leading to delays and stress. There may also be consequences for a space habitat design if squared volumes do not look square to astronauts. Virtual reality training may be a way to train the astronauts to compensate for such altered spatial representation. The purpose of this study is to investigate the effects of acute and long-term effects of exposure to microgravity on the mental representation of spatial cues by astronauts during and after space flight. The results of this study could have important consequences for human performance during space flight. For example, if an astronaut imagines a space station shorter than it is in reality, the expected duration of a trip within this station will be shorter than what it actually takes, thus leading to delays and stress. There may also be consequences for a space habitat design if squared volumes do not look square to astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Increased understanding of the effects of microgravity on astronauts during and after spaceflight.
Task Progress:	Research efforts conducted during this reporting period included experimental design, evaluation and testing of hardware, development of 3D software, support of initial testing during ground-based and parabolic flight studies, analysis of data collected during ground-based and parabolic flight studies, preparation of documents for Institutional Review Board (IRB) approval, and preparation of documents for Preliminary Design Review (PDR). Specific tasks performed are detailed below. Experimental Design: Reviewed/Refined Proposed ED and Protocols for Each Task (Handwriting/Drawing Tasks, Recognizing 2-D and 3-D Forms, Measuring Distance Perception), Coordinated experiment resource requirements, reviews, and schedules, ensuring that the following are met (Flight objectives, Committee for the Protection of Human Subjects protocols, Experiment procedures, Proper baseline data collection, Experimentation with unique software, and Experimentation with unique equipment development) Hardware Evaluation and Testing: Conducted comparative evaluation of the 6”x8” and 9”x 12” tablets, Conducted a comparative evaluation of the Cy-Visor, Cybermind, and eMagin 3D VR displays, Assessed current ISS laptop, including current software configuration, and Configured Sony laptop 3D Software Development: Reviewed Software Documents (Program Requirements Document LS-71000, HRF Software Development Plan Document LS-71020, Human-Computer Interface Design Guide LS-71130, Interface Definition Document LS-71042, Experiment Document Format LS-20079, Interface Design Document LS-71062), Identified Software Requirements based on the documents reviewed and experimental design specifications, Derived Software Specifications based on the requirements identified, and Designed Software Architecture (Selected 3D programming environment, Selected Digitizing Software, Completed GUI Design), and Began Software Programming Support of Initial Ground-Based and Parabolic Flight Testing: Analyzed Task 1 Experiment Data collected in Ground-Based and Parabolic Flight Experiments, Presented Preliminary Findings at Habitation Conference, and Conducted Task 2 Parabolic Flight Experiment (March, 2006) Preparation of Documents: Began Preparing Required IRB Documents, Prepared Required PDR Documents Administrative Tasks: Finalized Budget, Held Weekly Team Meetings, Coordinated with ESA PI, Completed Documentation and Reporting, and Visited Co-PI’s Laboratory
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Abstracts for Journals and Proceedings	Clément G, Lockerd A, Lathan C. "Mental Representation of Spatial Cues During Spaceflight (3D-SPACE)." Presented at the Habitation 2006 Conference, Orlando, FL, Feb 6-8, 2006. Habitation 2006;10(3-4):200-1. , Feb-2006